

Modelling Assignment 2 - Group Component

November 14, 2025

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[1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import scipy.stats as stats
```

1 Problem Statement

Construct a mathematical model of total annual income for the two locations of Loafe Café where daily average temperature, daily precipitation, and daily humidity affects the income and the total frequency of customers in the month of January 2025 and the month of July 2025.

2 Variables and Parameters

Description	Symbol	Dimension	Type
Total daily customers	C	1	Random Variable
Average daily temperature (°C)	T	Θ	Random Variable
Total daily precipitation (mm)	P	L	Random Variable
Average daily humidity (%)	H	1	Random Variable
Reference number of customers	C_0	1	Parameter
Reference temperature (°C)	T_0	Θ	Parameter
Reference precipitation (mm)	P_0	L	Parameter
Reference humidity (%)	H_0	1	Parameter
Temperature sensitivity	a	Θ^{-2}	Parameter
Precipitation sensitivity	b	L^{-1}	Parameter
Humidity sensitivity	c	1	Parameter
Average spending per customer (CAD)	$\bar{M}$	\$	Parameter
Total daily income (CAD)	M	\$	Random Variable
Total annual income (CAD)	M_{year}	\$	Random Variable

3 Assumptions and Constraints

- The total number of daily customers is the sum over both locations (Alumni and Sauder) and modelled by one function $C(T, P, H)$.
- Any other factors (i.e., locations of two cafés, student populations) besides weather have a negligible effect on the number of daily customers.

- Menu prices are fixed throughout the year and tips are not included in the income.
- The weather measurements are representative of UBC.
- The weather measurements from 1997 to 2024 also represent weather data in 2025.
- All customers on a given day experience the same daily mean of T , P , and H .
- Each day's weather is independent of the previous day.
- The weather condition variables T , P , and H are approximated by Gaussian KDEs.
- For each simulated year and month m , days are sampled independently from these KDEs.
- The customer parameters $(C_0, T_0, P_0, H_0, a, b, c, \bar{M})$ are constant throughout the year.
- We treat 2025 as a 365-day (non-leap) year.

4 Building Solutions

4.1 Model Equations

The total annual income is:

$$M_{\text{year}} = \sum_{t=1}^{365} M_t,$$

where

$$M_t = \bar{M} \cdot C, \quad C = \max\{c(T, P, H), 0\}, \quad c(T, P, H) = C_0 + a(T - T_0)^2 - b(P - P_0) - c(H - H_0)^2.$$

```
[2]: weather = pd.read_csv("weatherstats_vancouver_daily_2025.csv") # Import data
      ↪sets
customer = pd.read_csv("customer_data_2025.csv")

jan_total_cust = (customer["January_Frequency_Customers_Alumni_Centre"] +
      ↪customer["January_Frequency_Customers_Sauder"]).sum() # Extract total
      ↪customer in Jan
jul_total_cust = (customer["July_Frequency_Customers_Alumni_Centre"] +
      ↪customer["July_Frequency_Customers_Sauder"]).sum() # Extract total customer
      ↪in July
month_days = {1:31, 2:28, 3:31, 4:30, 5:31, 6:30, 7:31, 8:31, 9:30, 10:31, 11:
      ↪30, 12:31}

def monthly_kde(df, col): # return kde for every month with a given weather type
    return {m: stats.gaussian_kde(df.loc[df["Month"] == m, col].values) for m
      ↪in range(1,13)}

kde_T, kde_P, kde_H = monthly_kde(weather, "temperature"), monthly_kde(weather,
      ↪"precipitation"), monthly_kde(weather, "humidity")
```

4.2 Parameter Choice

We set $T_0 = 25^\circ\text{C}$ as the preferred indoor temperatures are $20\text{--}24^\circ\text{C}$ [Fanger, P. O., 1970; de Dear, R., 2017; ASHRAE, 2017]. Coastal Vancouver humidity of 70-90% and indoor comfort guidelines $\lesssim 60\%$ relative humidity (RH) suggest that $H_0 = 85\%$ [Fanger, P. O., 1970; Parsons, K., 2014;

ASHRAE, 2017]. Typical dry days have $P \approx 0$ mm and rainfall reduces each trip by about 10-30% [de Freitas, C. R., 2003; Koetse, M. J., 2009], so we set $P_0 = 0$ mm.

```
[3]: C0, T0, P0, H0, a, b, c = 280.0, 20.0, 0.0, 85.0, 0.50, 5.00, 0.50 #
      ↪Reference values

def C_hat(T,P,H): # given customer model
    return np.maximum(C0 + a*(T-T0)**2 - b*(P-P0) - c*(H-H0)**2, 0.0)

def month_expectation(m, n=100_000): # Monte Carlo expectation for a given month
    Tm, Pm, Hm = kde_T[m].resample(n), kde_P[m].resample(n), kde_H[m].
    ↪resample(n)
    return month_days[m] * np.mean(C_hat(Tm,Pm,Hm))

expected_jan_cust, expected_jul_cust = month_expectation(1),
    ↪month_expectation(7)

print(f"January 2025 (observed vs model): observed = {jan_total_cust:,.0f} |
    ↪expected = {expected_jan_cust:,.0f}")
print(f"July 2025 (observed vs model): observed = {jul_total_cust:,.0f} |
    ↪expected = {expected_jul_cust:,.0f}")
```

January 2025 (observed vs model): observed = 10,862 | expected = 10,684

July 2025 (observed vs model): observed = 5,438 | expected = 5,239

January and July 2025 total customer frequencies (10,862 and 5,438 customers, respectively) imply mean daily volume of about 350 and 175, so we set $C_0 = 280$ and sensitivities $a = 0.5, b = 5, c = 0.5$. This makes a 10°C , 10 mm, or 10% RH change shift demand by approximately 50 customers ($\approx 15\text{-}25\%$ of C_0), which is consistent with reported weather effects [de Freitas, C. R., 2003; Koetse, M. J., 2009]. We verify that these parameters reproduce the Janaury and July 2025 totals to within a few percent using Monte Carlo simulations based on month-wise KDE weather distributions.

```
[4]: # Annual Monte Carlo for total customers
N_years = 3000
annual_customers = np.zeros(N_years)

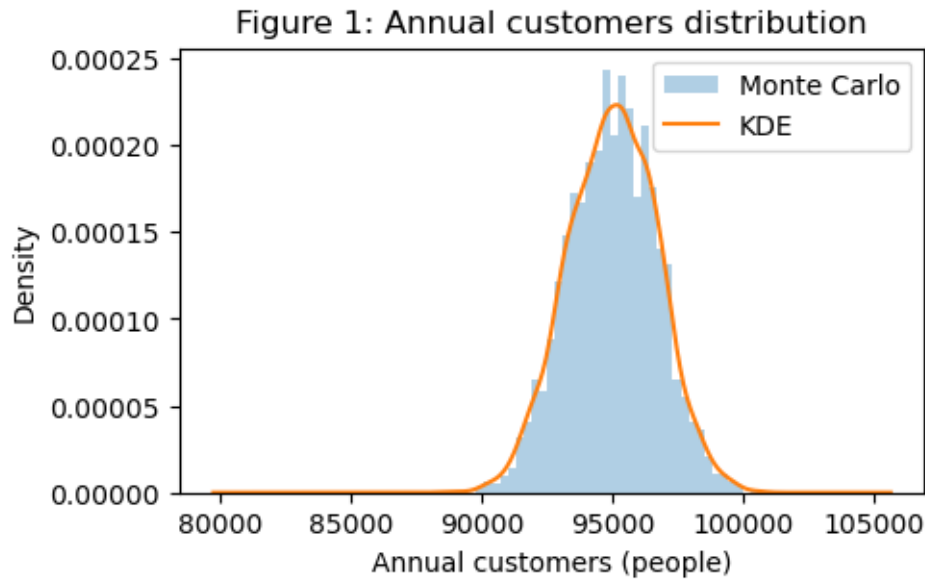
for month in range(1,13):
    days = month_days[month]
    Tm, Pm, Hm = kde_T[month].resample(days*N_years), kde_P[month].
    ↪resample(days*N_years), kde_H[month].resample(days*N_years)
    annual_customers += C_hat(Tm, Pm, Hm).reshape(N_years, days).sum(axis=1)

C_mean = np.mean(annual_customers)
C_med = np.median(annual_customers)
C_lo, C_hi = np.quantile(annual_customers, [0.025,0.975])

x = np.linspace(max(0, annual_customers.min()*0.9), annual_customers.max()*1.
    ↪05, 400)
```

```
plt.figure(figsize=(5,3))
plt.hist(annual_customers, bins=40, density=True, alpha=0.35, label="Monte Carlo")
plt.plot(x, stats.gaussian_kde(annual_customers)(x), label="KDE")
plt.xlabel("Annual customers (people)", plt.ylabel("Density"))
plt.title("Figure 1: Annual customers distribution")
plt.legend(), plt.show()

print(f"Annual customers (Monte Carlo): Mean = {C_mean:,.0f} | Median = {C_med:,.0f} | 95% CI = [{C_lo:,.0f},{C_hi:,.0f}]")
```



Annual customers (Monte Carlo): Mean = 94,998 | Median = 95,040 | 95% CI = [91,664,98,255]

4.3 Annual Income Scenarios

Google Maps reports a typical spending range of \$10-20 per customer at both locations of Loafe Café [Google Maps, 2025]. Based on this, we consider three fixed average spending values $\bar{M} \in \{\$10.00, \$15.00, \$20.00\}$.

```
[5]: spend_scenarios = [10.00, 15.00, 20.00]

plt.figure(figsize=(5,3))
for mbar in spend_scenarios:
    annual_income_fixed = mbar*annual_customers
    kde_inc = stats.gaussian_kde(annual_income_fixed)
    x_fix = np.linspace(annual_income_fixed.min()*0.95, annual_income_fixed.
        max()*1.05, 400)
```

```

plt.plot(x_fix, kde_inc(x_fix), label=f"fixed M = ${mbar:.2f}")

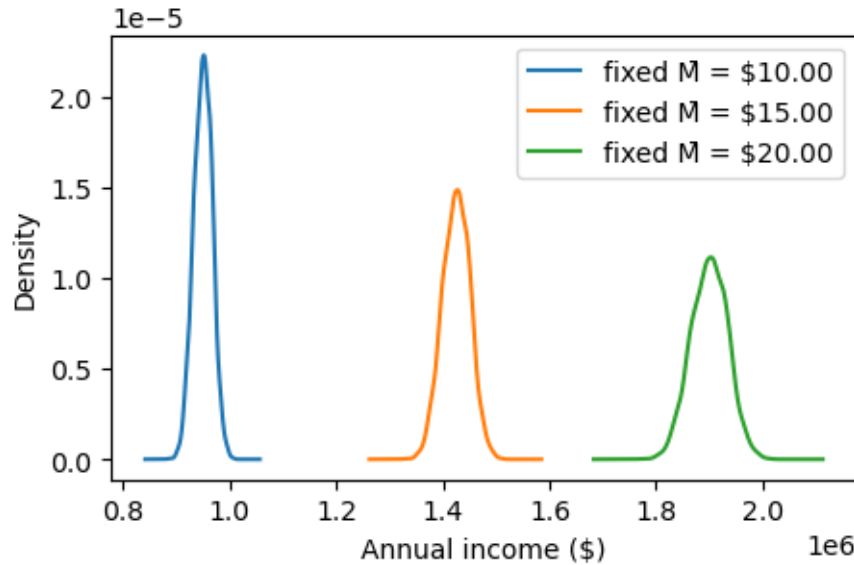
plt.xlabel("Annual income ($)"), plt.ylabel("Density")
plt.title("Figure 2: Annual income - fixed vs stochastic average spending")
plt.legend(), plt.show()

print("Annual income summary by fixed-spend scenario:")

for mbar in spend_scenarios:
    annual_income_fixed = mbar * annual_customers
    mu = np.mean(annual_income_fixed)
    lo, hi = np.quantile(annual_income_fixed, [0.025, 0.975])
    print(f"M = ${mbar:>4.2f}: mean = ${mu:,.0f}, 95% CI = [{lo:,.0f}, {hi:,.0f}]")

```

Figure 2: Annual income - fixed vs stochastic average spending



Annual income summary by fixed-spend scenario:

M = \$10.00: mean = \$949,985, 95% CI = [\$916,637, \$982,552]

M = \$15.00: mean = \$1,424,977, 95% CI = [\$1,374,955, \$1,473,828]

M = \$20.00: mean = \$1,899,970, 95% CI = [\$1,833,274, \$1,965,104]

5 Analyze and Assess

5.1 Observations and Analysis

We sampled the annual total customers through Monte Carlo. This is represented by the blue histogram in the figure above. We simulated 3000 possible years taking into account the weather conditions and calculated customer count per day accordingly. Then we summed them across the

days to get the annual total customer count. We also fitted the KDE to estimate the distribution of the annual customers.

Overall, our plot is shaped like a bell curve since some simulated years would have ‘better’ weather (T, P, H that deviate less from reference values) therefore more customers, and some simulated years had ‘worse’/less ideal weather so they have fewer customers. Due to the central limit theorem, the total tends to approximately take the shape of a normal distribution, with a mean of about 94,998 customers per year. Having a normal distribution suggests that, in our model, Loafe Café has a relatively stable annual customer volume when random fluctuations in weather conditions are taken into account. We can see that the data mostly lies between 90,000 and 100,000 customers and that the peak of the histogram is at around 95,000 customers per year which means it is the most likely outcome, the width shows the uncertainty due to weather randomness (Figure 1).

After multiplying (scaling) our annual customer distribution by $\bar{M}$ (average spending per customer), we obtain a uni-modal (approximately normal) distribution (Figure 2). Each curve represents a different $\bar{M}$ value. The shape and spread are pretty similar across different average spending amounts which tells us that the weather affects customers in similar ways.

5.2 Model Limitations

We assumed that all other factors besides weather have a negligible effect on the number of daily customers to simplify our model. Realistically, there are many other parameters affecting the number of customers. For example, the customer counts in January and July would vary as there is a larger number of students on campus at the beginning of the winter session as opposed to July, which is during the summer session. Including variables like the number of students on campus (taking into consideration the time in the semester and exam seasons) would strengthen our model and help us make more accurate predictions about the income. The income per day could also vary with promotions and popularity of seasonal drinks; including this in our model could account for that variability.

Usually, businesses do not have fixed menu prices in the long run. Prices can be affected by the demand for their product, competition, and supply constraints, assuming fixed menu prices in turn assume a very simple economic model that can be improved by considering economic factors.

Our assumption that the weather data from 1997 to 2024 is representative of weather data in 2025 can also be improved by either collecting or generating 2025 weather data, assuming historical data is representative disregards weather trends such as global warming and climate change.

Assuming that the average daily spending per customer is fixed also limits our model, as we do not account for variables such as different types of customers (students/staff/other) and different income levels.

6 Bibliography

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